Subseasonal-to-Seasonal Forecasting of Drought-to-Flood Transitions

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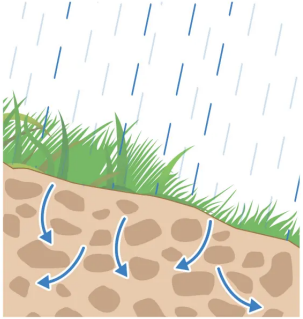
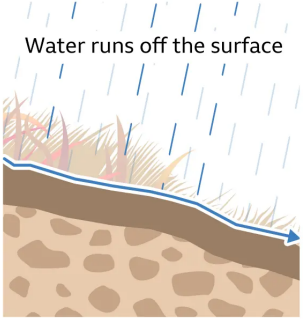
What Is a Drought-to-Flood Transition?

Normal conditions



Soil absorbs water like a sponge

Drought conditions



Hard layer of soil repels water

Adapted from BBC Research (2022)

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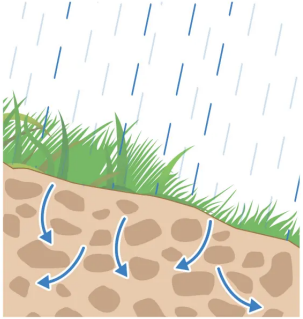

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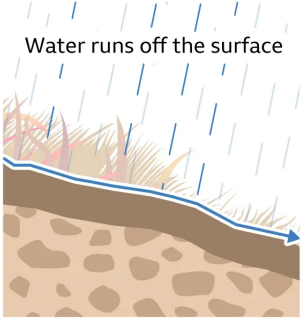

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Matanó et al. (2024)

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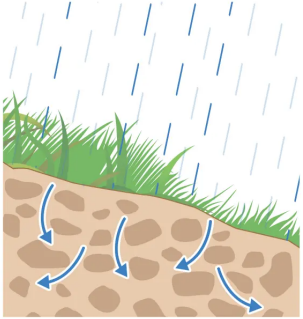

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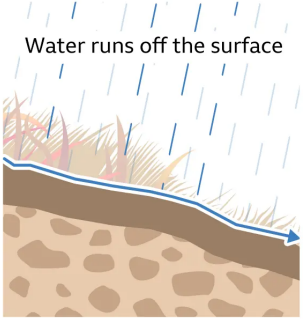

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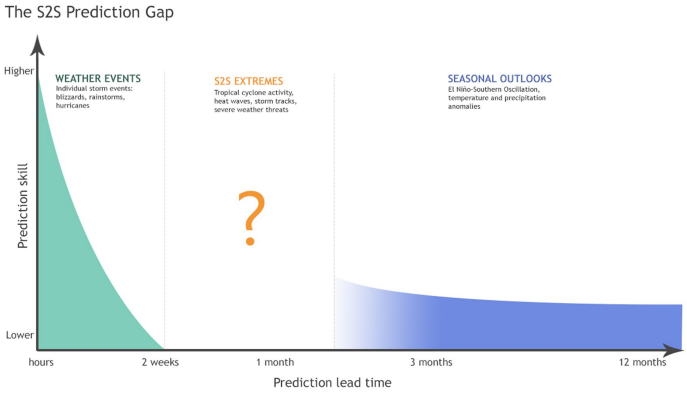
~6 weeks

average transition time

Yang et al. (2025)

Why Is This So Hard to Forecast?



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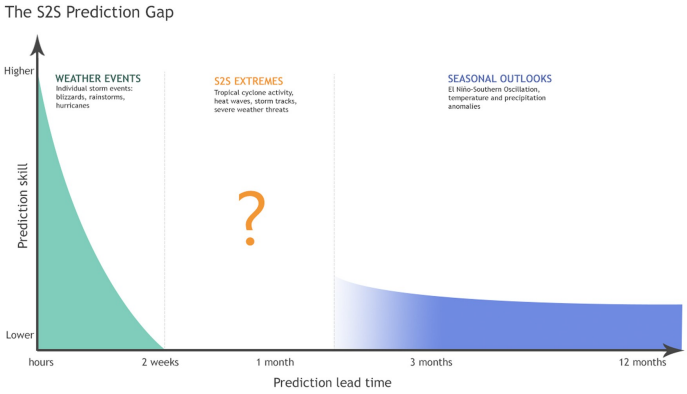

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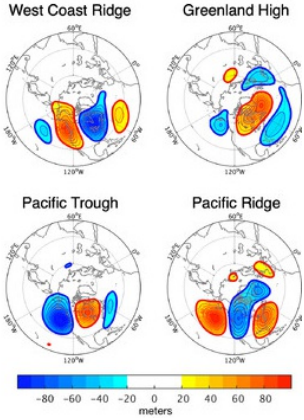
Weather regimes



Adapted from DeFlorio et al. (2024)

What Causes These Transitions?

Persistent blocking

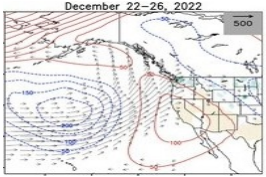
Rainfall suppressed
Soil dries out

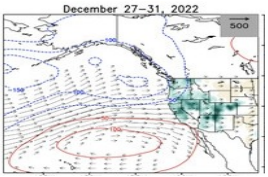
Blocking breaks down

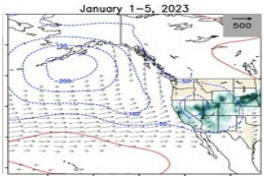
Pattern shifts
Rain arrives

Transition

Rain hits dry soil
Runoff amplified







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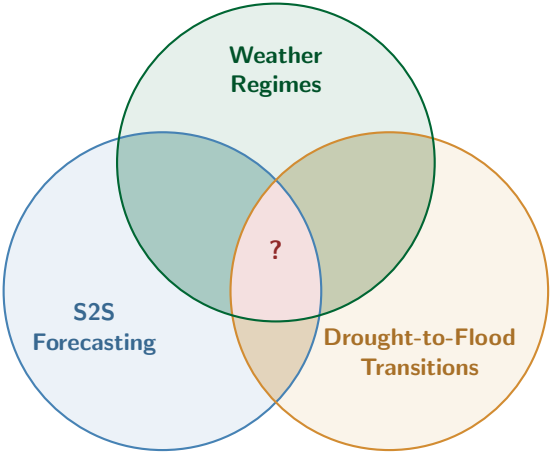
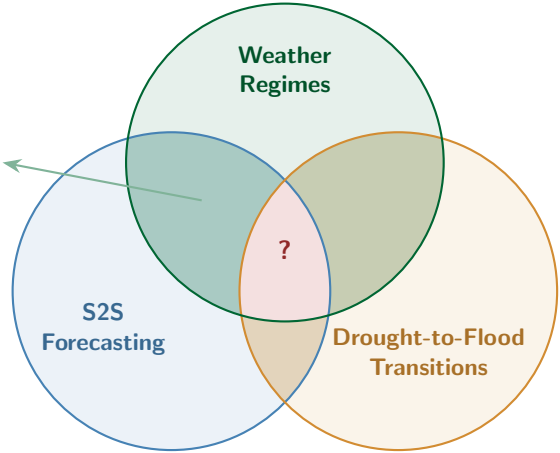


Where Does That Leave Us?

✓ Regimes are predictable at S2S timescales



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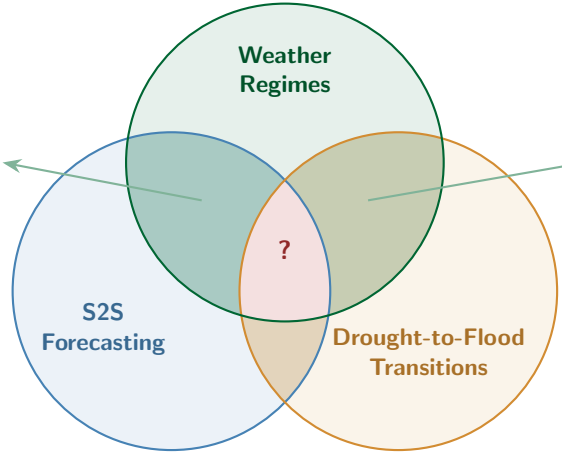

✓ Regimes drive these transitions

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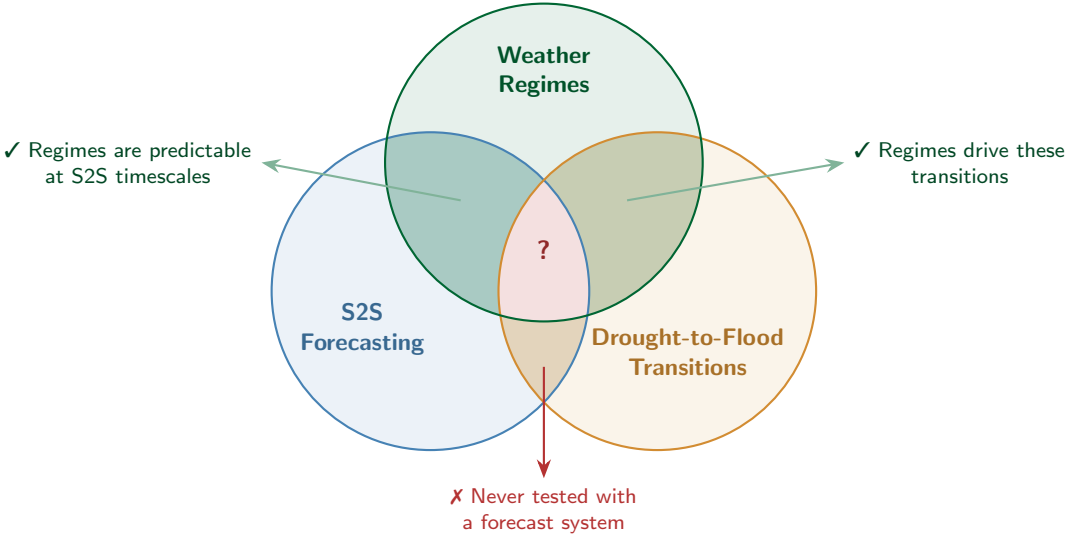

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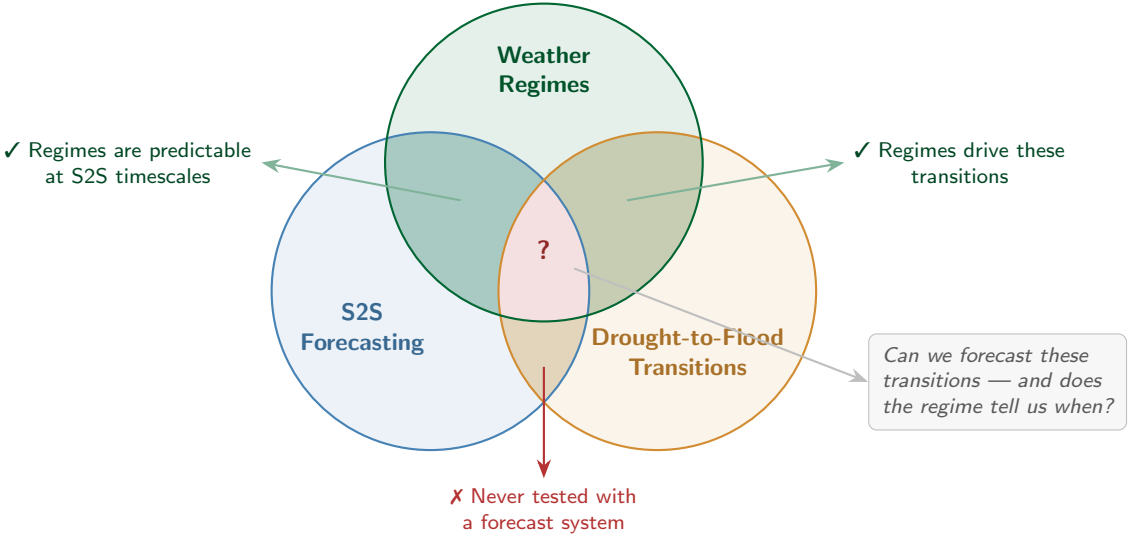
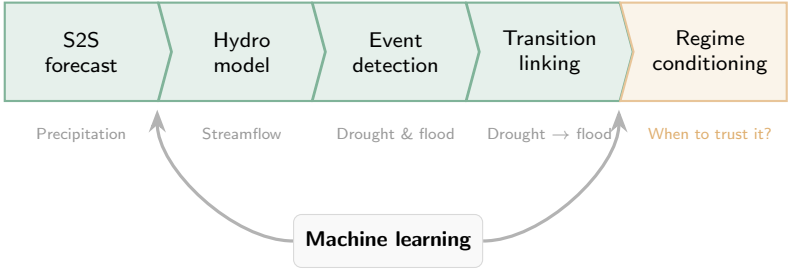
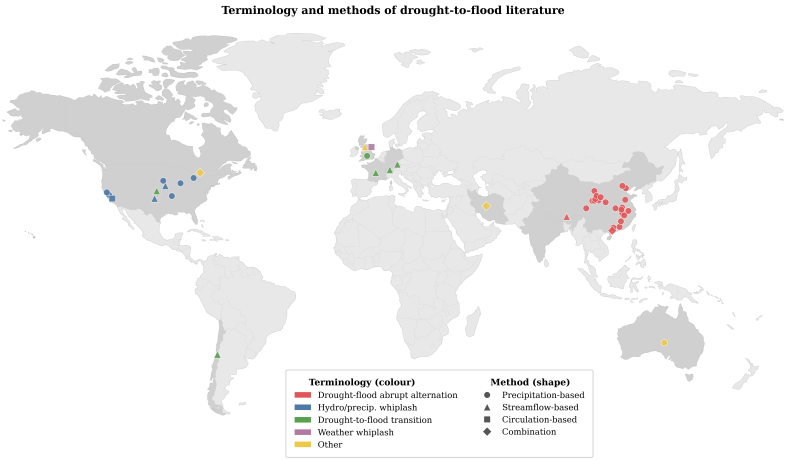


How Do I Test This?



Catchment in drought.
Blocking breaking down weeks 3–4.
Above-normal transition risk weeks 4–6.

How Do We Detect These Events?



5+ terms

for the same phenomenon across different traditions

Opposite trends

rainfall-based vs streamflow-based detection over the same period

Yang et al. (2025)

5 of 8 missed

known impactful events undetected by standard methods

Anderson et al. (2025)

Takeaways

1. Drought-to-flood transitions are common, damaging, and play out over the S2S window

2. The regime shifts that drive them are predictable at this range

3. Nobody has tried to forecast drought-to-flood transitions. That's what I hope to do

References

- DeFlorio, M. J., and Coauthors, 2024: From California's extreme drought to major flooding: Evaluating and synthesizing experimental seasonal and subseasonal forecasts during winter 2022/23. *Bull. Amer. Meteor. Soc.*, **105**, E84–E104.
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